

Group members:

1. ???
2. ???

SECJ2013 - DSA

2024/2025-1

QUIZ 2

ALGORITHM EFFICIENCY

Determine the big notation of the following expressions:

A. $2n + 5n^2 \Rightarrow O(n^2)$

B. $3n \log_2 n + 5n - 10n + 5n \Rightarrow O(n \log_2 n)$

C. $3 + \log_2 n + 2n + n^2 + 2^n \Rightarrow O(2^n)$

D. $1 + 2 + 3 \Rightarrow O(1)$

E. $1n - 2n + 1n \Rightarrow O(n)$

List the order of the above expression from the most simple to the complex one:

$D < E < B < A < C$

ALGORITHM EFFICIENCY

Calculate the number of steps and determine the big-O notation of the following program:

Statement	Number of Steps
<code>int start = 0, stop = 10;</code>	1
<code>for (int i = start; i <= stop; i++) {</code>	n
<code> cout << "Hello\n";</code>	n .1
<code>}</code>	0
Total steps	$1+n+n+0 = 1 + 2n$
Big-O notation	O(n)

ALGORITHM EFFICIENCY

Calculate the number of steps and determine the big-O notation of the following program:

Statement	Number of Steps
<code>int i, j, n = 8;</code>	1
<code>for (i = 1; i < n; i++) {</code>	n-1
<code>j = 1;</code>	(n-1) .1
<code>while (j <= i) {</code>	n.(n-1)/2
<code>j++;</code>	(n.(n-1)/2) .1
<code>cout << i + j << "\n";</code>	(n.(n-1)/2) .1
<code>}</code>	0
<code>}</code>	0
Total steps	$1 + (n-1) + (n-1) + (n.(n-1)/2) + (n.(n-1)/2) + (n.(n-1)/2) + 0 + 0 = 2n - 1 + 3n^2 - n/2$
Big-O notation	$O(n^2)$

SIMPLE SORT

1. Given that `num` array has been defined as follows:

```
int num = [ 2, 1, 4, 3, 5];
```

List out the content of the `num` array after passes 1 and 2 if it was sorted by using bubble, selection, and insertion sort in ascending order.

Passes	Bubble Sort	Selection Sort	Insertion Sort
After Pass 1	1, 2, 3, 4, 5	2, 1, 4, 3, 5	1, 2, 4, 3, 5
After Pass 2	1, 2, 3, 4, 5	2, 1, 3, 4, 5	1, 2, 4, 3, 5

2. For an array `[4, 3, 1, 2]`, determine the type of sorting (ascending/descending) and the implemented algorithm (bubble/improved bubble/selection/insertion) for the following output:

Pass1: 4 3 2 1

Pass2: 4 3 2 1

Num. of comparisons: 5

Num. of swaps: 1

Answer: Descending, Improved Bubble Sort

ADVANCE SORT

Given two halves of array to be merged in the last merge operation as follows:

[1, 7, 9,] [2, 4, 6, 10]

Complete the tracing of the value of first1, first2, value comparison, and the temporary array content until all items are fully copied and sorted into the temporary array:

first1	first2	value comparison	tempArray
0	3	1 < 2	[1, null, null, null, null, null, null]
1	3	7 > 2	[1, 2, null, null, null, null, null]
1	4	7 > 4	[1, 2, 4, null, null, null, null]
1	5	7 > 6	[1, 2, 4, 6, null, null, null]
1	6	7 < 10	[1, 2, 4, 6, 7, null, null]
2	6	9 < 10	[1, 2, 4, 6, 7, 9, 10]

ADVANCE SORT

For an array `int list = [12, 6, 16, 4, 14, 2]`, with pivot value is 12 (the first element), trace the position of the bottom (**B**) and top (**T**) index and list the content of the array after each 1st, 2nd and 3rd swapping operation in the `partition(list, 0, 5)` function of a quick sort algorithm (ascending order).

Initial position => [12, 6, 16, 4, 14, 2]
 B **T**

After 1st swap => [2, 6, 16, 4, 14, 12]
 B **T**

After 2nd swap => [2, 6, 12, 4, 14, 16]
 B **T**

After 3rd swap => [2, 6, 4, 12, 14, 16]
 BT

What is the cut point returned by the `partition(list, 0, 5)` function and the next two calls of quick sort function: `quickSort(list, ?, ?)`?

cut point => 3
`partition(list, 0, 3)`
`partition(list, 4, 5)`